Supervised Classification: HCC Tumor vs Matched Normal Liver

Team

Name	Email	GitHub
Junhao Li	lijunhao@ufl.edu	https://github.com/JunhaoLiXD

Dataset

RNA-seq of 50 Paired Hepatocellular Carcinoma Samples

Source: https://www.refine.bio/experiments/SRP068976/rna-seq-of-50-paired-hepatocellular-carcinoma?ref=search

Scientific Question

Can gene expression profiles distinguish hepatocellular carcinoma (HCC) tumor tissue from matched non-tumor liver tissue?

GitHub Repository

https://github.com/JunhaoLiXD/HCC-RNASeq-Classification

1. Load the expression data and metadata

We reused the expression matrix and sample groups processed in Assignment 2 (100 samples from 50 patients: 50 tumor and 50 matched normal) and our own K-means cluster assignments from Assignment 3. Genes with more than 10 counts in at least 10 samples were kept (23,393 genes) and values were log2(count + 1) transformed. The patient number in each sample title (e.g. “Patient 39 Tumor”) was kept as a grouping variable, so that the tumor and normal sample of the same patient are never split between training and test data.

2. Supervised analysis

As a team of one, two methods from the list were run in Python with scikit-learn: Method 1 – logistic regression (default L2 penalty, C = 1.0, lbfgs solver) and Method 2 – random forest (500 trees, max_features = $\sqrt{p}$, random_state = 42). Each model is a pipeline that selects the N most variable genes, z-scores them and fits the classifier; gene selection and scaling are refitted inside every training fold, so the held-out samples never influence the features. All main results use the 5,000 most variable genes.

Performance was estimated with 5-fold cross-validation that is stratified by class and grouped by patient (StratifiedGroupKFold), repeated 10 times with different random seeds. Every repeat gives one out-of-fold prediction for every sample; AUC is reported as mean

± SD over the 10 repeats, and accuracy as the mean. Grouping by patient matters for this paired design: if a patient's normal sample were in the training data and the tumor sample in the test data, the model could partly recognise the individual rather than the tissue type.

Each method was trained on three targets: (1) the Assignment 1 groups, Tumor vs Normal; (2) the Assignment 3 K-means clusters at k = 2 (53 / 47 samples); and (3) the K-means clusters at k = 3 (51 / 36 / 13 samples).

2.1 Predicting Tumor vs Normal

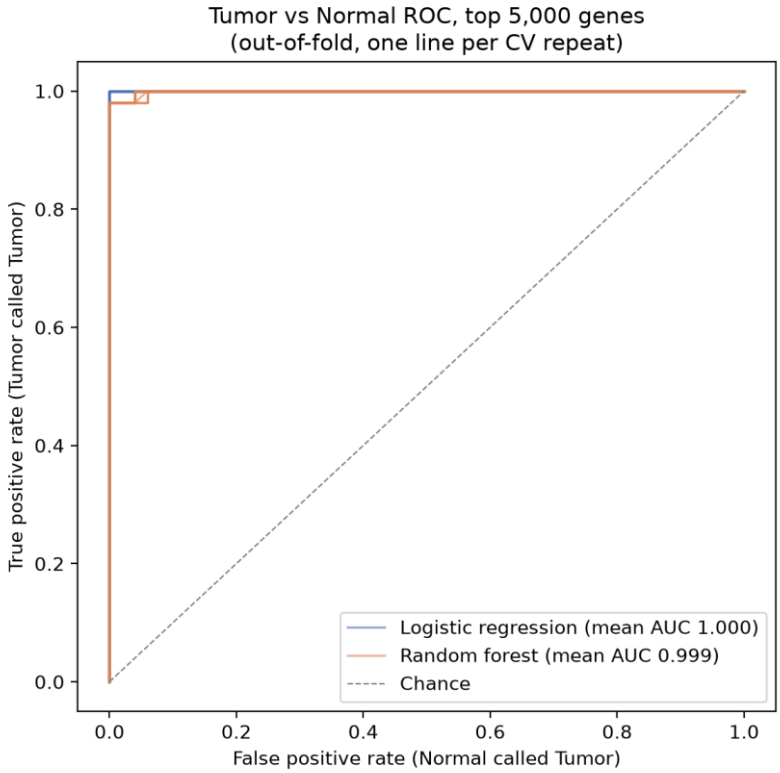


Figure 1. Out-of-fold ROC curves for Tumor vs Normal, top 5,000 genes; one line per CV repeat (10 per method).

Figure 1 shows the out-of-fold ROC curves of both models for Tumor vs Normal with the 5,000 most variable genes, one curve per CV repeat. Logistic regression reached a mean AUC of 1.000 (SD < 0.001) and random forest 0.999 (SD < 0.001); both curves rise almost vertically, so nearly every tumor receives a higher tumor probability than every normal sample. At a 0.5 threshold the accuracy is 0.990 for logistic regression and 0.989 for random forest, i.e. about one sample in 100 is misclassified. That sample is Patient 39 Tumor, which both models call Normal in every repeat – the same tumor sample that clustered with normal liver in Assignment 3.

2.2 Predicting the Assignment 3 clusters

We retrained both methods to predict our own K-means clusters from Assignment 3. K-means was used rather than hierarchical clustering because, at k = 3, it splits the tumors into two sizeable subgroups (36 and 13 samples), whereas average-linkage clustering tends to split off single outlying samples at higher k. Both logistic regression (multinomial) and random forest handle more than two classes directly, so k = 3 was fitted as one multiclass model instead of separate 1-vs-rest binary models; this gives one probability per cluster for every sample, and the AUC for k = 3 is the macro-averaged one-vs-rest AUC. k = 2 was kept as well because it is almost identical to Tumor / Normal and serves as a reference point.

With 5,000 genes the k = 2 clusters are learned as well as the biological groups (AUC 1.000 for logistic regression and 1.000 for random forest; accuracy 0.989 and 0.982). The k = 3 clusters are harder: the macro AUC is still high (0.997 and 0.998), but accuracy drops to 0.967 and 0.953 because the two tumor subclusters are not sharply separated. Because the cluster labels were themselves derived from the same expression data, these numbers measure how reproducibly the unsupervised partition can be learned from held-out samples, not an independent biological validation.

2.3 Sample-specific predictions across models

To compare models sample by sample, every combination of method and number of genes (2 methods × 10, 100, 1,000, 5,000 and 10,000 genes) was treated as one model, giving 10 models per target. A model's label for a sample is the class with the highest out-of-fold probability averaged over the 10 CV repeats. This produced a matrix of 100 samples × 30 models in which each cell is the label a model assigned to a sample. As a sample-level counterpart of AUC we also recorded, for every sample, the mean probability the models assign to its true label (1 = always confidently correct).

Table 1. Samples × models matrix of predicted labels (partial: 12 of the 100 samples; the full matrix for all 100 samples is sample_by_model.csv in the GitHub repository). Columns are models (LR = logistic regression, RF = random forest, followed by the number of genes); "True" is the sample's group or own K-means cluster, and highlighted cells differ from it.

Sample	True	LR 10	LR 100	LR 1k	LR 5k	LR 10k	RF 10	RF 100	RF 1k	RF 5k	RF 10k
Tumor vs Normal models (true label = Assignment 1 group)											
P1 Normal	Normal	Normal	Normal	Normal	Normal	Normal	Normal	Normal	Normal	Normal	Normal
P2 Normal	Normal	Normal	Normal	Normal	Normal	Normal	Normal	Normal	Normal	Normal	Normal
P25 Normal	Normal	Normal	Normal	Normal	Normal	Normal	**Tumor**	Normal	Normal	Normal	Normal
P1 Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor
P2 Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor
P6 Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor
P18 Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor
P23 Tumor	Tumor	**Normal**	Tumor	Tumor	Tumor	Tumor	**Normal**	**Normal**	Tumor	Tumor	Tumor
P25 Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor
P39 Tumor	Tumor	**Normal**	**Normal**	**Normal**	**Normal**	**Normal**	**Normal**	**Normal**	**Normal**	**Normal**	**Normal**
P45 Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor
P47 Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor	Tumor
K-means k = 2 models (true label = own cluster)											
P1 Normal	1	1	1	1	1	1	1	1	1	1	1
P2 Normal	1	1	1	1	1	1	1	1	1	1	1
P25 Normal	1	1	1	1	1	1	**2**	1	1	1	1
P1 Tumor	1	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**
P2 Tumor	1	**2**	**2**	1	1	1	**2**	**2**	**2**	**2**	**2**
P6 Tumor	2	2	2	2	2	2	2	2	2	2	2
P18 Tumor	2	2	2	2	2	2	2	2	2	2	2
P23 Tumor	2	**1**	**1**	2	2	2	**1**	**1**	2	2	2
P25 Tumor	2	2	2	2	2	2	2	2	2	2	2
P39 Tumor	1	1	1	1	1	1	1	1	1	1	1
P45 Tumor	2	2	2	2	2	2	2	2	2	2	2
P47 Tumor	2	2	2	2	2	2	2	2	2	2	2
K-means k = 3 models (true label = own cluster)											
P1 Normal	1	1	1	1	1	1	1	1	1	1	1
P2 Normal	1	1	1	1	1	1	1	1	1	1	1
P25 Normal	1	**2**	1	1	1	1	**2**	1	1	1	1
P1 Tumor	2	2	2	2	2	2	2	2	2	2	2
P2 Tumor	2	2	2	2	2	2	2	2	2	2	2
P6 Tumor	2	2	2	2	2	2	2	2	2	2	2
P18 Tumor	3	**2**	**2**	**2**	**2**	3	**2**	**2**	**2**	**2**	**2**
P23 Tumor	2	**1**	**1**	2	2	2	**1**	**1**	2	2	2
P25 Tumor	3	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**
P39 Tumor	1	1	1	1	1	1	1	1	1	1	1
P45 Tumor	3	**2**	3	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**
P47 Tumor	3	**2**	3	3	3	3	**2**	**2**	**2**	3	3

Table 1 shows part of the samples × models matrix: 12 samples, chosen to include typical samples (e.g. Patient 1 Normal and Patient 6 Tumor) and the samples on which the models disagree, with one row per sample for each of the three targets. Across all 100 samples, 74 are predicted correctly by all 30 models, and the errors follow clear patterns: Patient 39 Tumor is called Normal by every Tumor / Normal model, Patient 1 Tumor is placed in tumor cluster 2 by every k = 2 model although K-means put it in cluster 1, and Patient 25

Tumor is assigned to the other tumor subcluster by every k = 3 model. Errors are most frequent in the 10-gene models (LR 10 and RF 10) and in the k = 3 block, which already hints at the gene-number effect in Section 2.5.

Table 2. Number of models (out of 10 per target) predicting each label, for the 26 samples that are not predicted correctly by every model. "Own" is the sample's K-means cluster from Assignment 3; C1–C3 are cluster labels.

Sample	Tumor	Normal	k=2 own	k=2: C1	k=2: C2	k=3 own	k=3: C1	k=3: C2	k=3: C3
P25 Normal	1	9	1	9	1	1	8	2	0
P1 Tumor	10	0	1	0	10	2	0	10	0
P2 Tumor	10	0	1	3	7	2	0	10	0
P3 Tumor	10	0	2	2	8	2	2	8	0
P4 Tumor	10	0	2	0	10	3	0	3	7
P5 Tumor	10	0	2	0	10	2	0	9	1
P11 Tumor	10	0	2	0	10	2	0	9	1
P13 Tumor	10	0	2	0	10	2	0	9	1
P18 Tumor	10	0	2	0	10	3	0	9	1
P19 Tumor	10	0	2	0	10	2	0	9	1
P23 Tumor	7	3	2	4	6	2	4	6	0
P24 Tumor	10	0	2	0	10	3	0	1	9
P25 Tumor	10	0	2	0	10	3	0	10	0
P26 Tumor	10	0	2	0	10	2	0	9	1
P28 Tumor	10	0	2	0	10	3	0	2	8
P29 Tumor	10	0	2	0	10	3	0	1	9
P32 Tumor	10	0	2	0	10	3	0	2	8
P38 Tumor	10	0	2	0	10	3	0	4	6
P39 Tumor	0	10	1	10	0	1	10	0	0
P40 Tumor	10	0	2	0	10	3	0	6	4
P41 Tumor	10	0	2	0	10	3	0	1	9
P42 Tumor	10	0	2	0	10	2	0	8	2
P45 Tumor	10	0	2	0	10	3	0	9	1
P47 Tumor	10	0	2	0	10	3	0	4	6
P48 Tumor	10	0	2	0	10	3	0	2	8
P50 Tumor	10	0	2	0	10	2	0	8	2

How many models predict each class label, for that sample?

Table 2 counts, for the 26 samples not predicted correctly by every model, how many of the 10 models per target assign each label; the other 74 samples receive their true label from all 30 models. For 97 of the 100 samples all 10 Tumor / Normal models predict the true group. The exceptions are Patient 39 Tumor (0 of 10 models predict Tumor), Patient 23 Tumor (7 Tumor, 3 Normal) and Patient 25 Normal (9 Normal, 1 Tumor). Patient 39 Tumor is therefore not a borderline case but a tumor whose expression profile looks like normal liver to every model; its mean probability for the Tumor label is only 0.17.

How many models predict the same cluster, for that sample?

At k = 2 all 10 models agree on one cluster for 96 samples, and for 95 of them that cluster is the sample's own K-means cluster. The main disagreement is with K-means itself: Patient 1 Tumor and Patient 2 Tumor were placed in the normal-like cluster 1 in Assignment 3, but 10 and 7 of 10 models predict the tumor cluster 2 for them, so their K-means assignments appear unstable. At k = 3 all 10 models agree for 78 samples (77 matching the own cluster); the remaining disagreements are almost all tumors split between clusters 2 and 3, and three tumors (Patients 18, 25 and 45) are predicted to the other tumor subcluster by at least 9 of 10 models.

2.4 Stability of class-label vs cluster predictions

We measured the stability of each sample's predictions in two ways: model agreement (the share of the 10 models that give the most common label) and the mean probability of the true label across the 10 models. The stability of the Tumor / Normal prediction was compared with the stability of the cluster prediction for k = 2 and k = 3 using Spearman correlation on both measures and Fisher's exact test on a 2 × 2 table of "all 10 models agree" (yes / no). The six p-values were adjusted together with the Benjamini–Hochberg method (Table 3).

Table 3. Tests of whether Tumor / Normal prediction stability correlates with cluster prediction stability (n = 100 samples; padj = BH-adjusted over all six tests).

Cluster target	Test	Stability measure	Statistic	p	padj
K-means k = 2	Spearman	Model agreement	ρ = 0.70	5.4e-16	1.6e-15
K-means k = 2	Spearman	Mean true-label probability	ρ = 0.78	9.2e-22	5.5e-21
K-means k = 2	Fisher's exact	All 10 models agree (yes/no)	∞ (odds ratio)	0.0012	0.0024
K-means k = 3	Spearman	Model agreement	ρ = 0.31	0.002	0.0029
K-means k = 3	Spearman	Mean true-label probability	ρ = 0.16	0.1	0.1
K-means k = 3	Fisher's exact	All 10 models agree (yes/no)	∞ (odds ratio)	0.047	0.056

Does the stability of the cluster and class label prediction correlate?

For k = 2, yes: both Spearman correlations are strong and significant after correction (ρ = 0.70 for model agreement, padj = 1.6e-15; ρ = 0.78 for mean true-label probability, padj = 5.5e-21), and Fisher's test is also significant (padj = 0.0024) because the only two samples with unstable group predictions (Patient 23 Tumor and Patient 25 Normal) are also unstable for the cluster. For k = 3 the association is weak: agreement still correlates modestly (ρ = 0.31, padj = 0.0029), but the probability-based correlation (ρ = 0.16, padj = 0.10) and Fisher's test (padj = 0.056) are not significant. The k = 2 clusters essentially encode Tumor / Normal, so both predictions fail on the same samples, whereas the k = 3 clusters add a split within the tumors that is unstable for different samples than the Tumor / Normal boundary. Most samples have agreement = 1, so the agreement-based tests rest on few informative samples; the probability-based measure is the more sensitive of the two.

2.5 Effect of the number of genes

We retrained every model on the 10, 100, 1,000, 5,000 and 10,000 most variable genes, with the same pipeline and cross-validation (gene selection inside each fold).

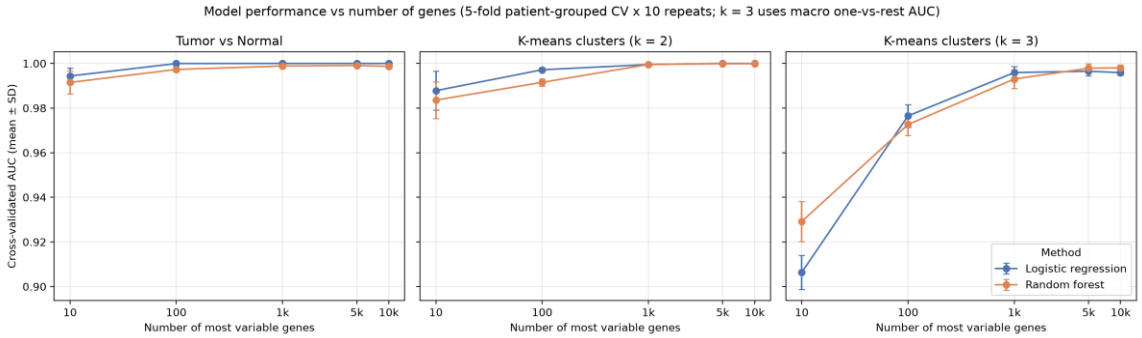


Figure 2. Cross-validated AUC (mean ± SD over 10 repeats) vs number of genes for Tumor vs Normal (left), K-means k = 2 (middle) and K-means k = 3 (right; macro one-vs-rest AUC).

Figure 2 plots the cross-validated AUC of both methods against the number of genes (log scale), with one panel per target and error bars showing the SD over the 10 CV repeats. For Tumor vs Normal and k = 2 the curves start high and flatten almost immediately, while for k = 3 they rise steeply from 10 to 1,000 genes. Random forest is ahead of logistic regression at 10 genes for k = 3 (0.929 vs 0.906), but the methods are within 0.005 of each other from 1,000 genes upwards.

Table 4. Cross-validated AUC (mean ± SD over 10 repeats) for every model and number of genes. T/N = Tumor vs Normal, LR = logistic regression, RF = random forest.

Genes	T/N LR	T/N RF	k=2 LR	k=2 RF	k=3 LR	k=3 RF
10	0.994 ± 0.004	0.991 ± 0.005	0.988 ± 0.009	0.984 ± 0.008	0.906 ± 0.008	0.929 ± 0.009
100	1.000 ± 0.000	0.997 ± 0.001	0.997 ± 0.001	0.992 ± 0.002	0.976 ± 0.005	0.973 ± 0.005
1,000	1.000 ± 0.000	0.999 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.996 ± 0.003	0.993 ± 0.004
5,000	1.000 ± 0.000	0.999 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.997 ± 0.002	0.998 ± 0.002
10,000	1.000 ± 0.000	0.999 ± 0.001	1.000 ± 0.000	1.000 ± 0.000	0.996 ± 0.001	0.998 ± 0.001

How did the number of genes affect the results?

More genes help, but only up to about 1,000 genes. Tumor vs Normal is already well predicted with 10 genes (AUC 0.994 and 0.991) and reaches 0.997–1.000 from 100 genes on. The k = 3 clusters depend much more on the number of genes: accuracy (model_performance.csv in the repository) rises from 0.83–0.84 with 10 genes to 0.95–0.97 with 5,000 genes, because the 10 most variable genes (which include sex-linked genes such as XIST, Assignment 3) carry little information about the tumor subclusters.

Does the AUC increase or decrease as the number of genes changes?

Table 4 lists the cross-validated AUC (mean and SD over the 10 repeats) for all 30 models. AUC increases with the number of genes for every model and target until it saturates at 1,000–5,000 genes. Beyond that point the changes are within the SD across CV repeats and slightly negative in some cases (e.g. Tumor/Normal random forest 0.9991 at 5,000 vs 0.9987 at 10,000 genes; k = 3 logistic regression 0.9965 vs 0.9959), consistent with the extra low-variance genes adding noise rather than signal. The largest gain is for k = 3, from 0.906 / 0.929 (10 genes) to 0.996 / 0.998 (10,000 genes).

3. Heatmaps and dendrograms

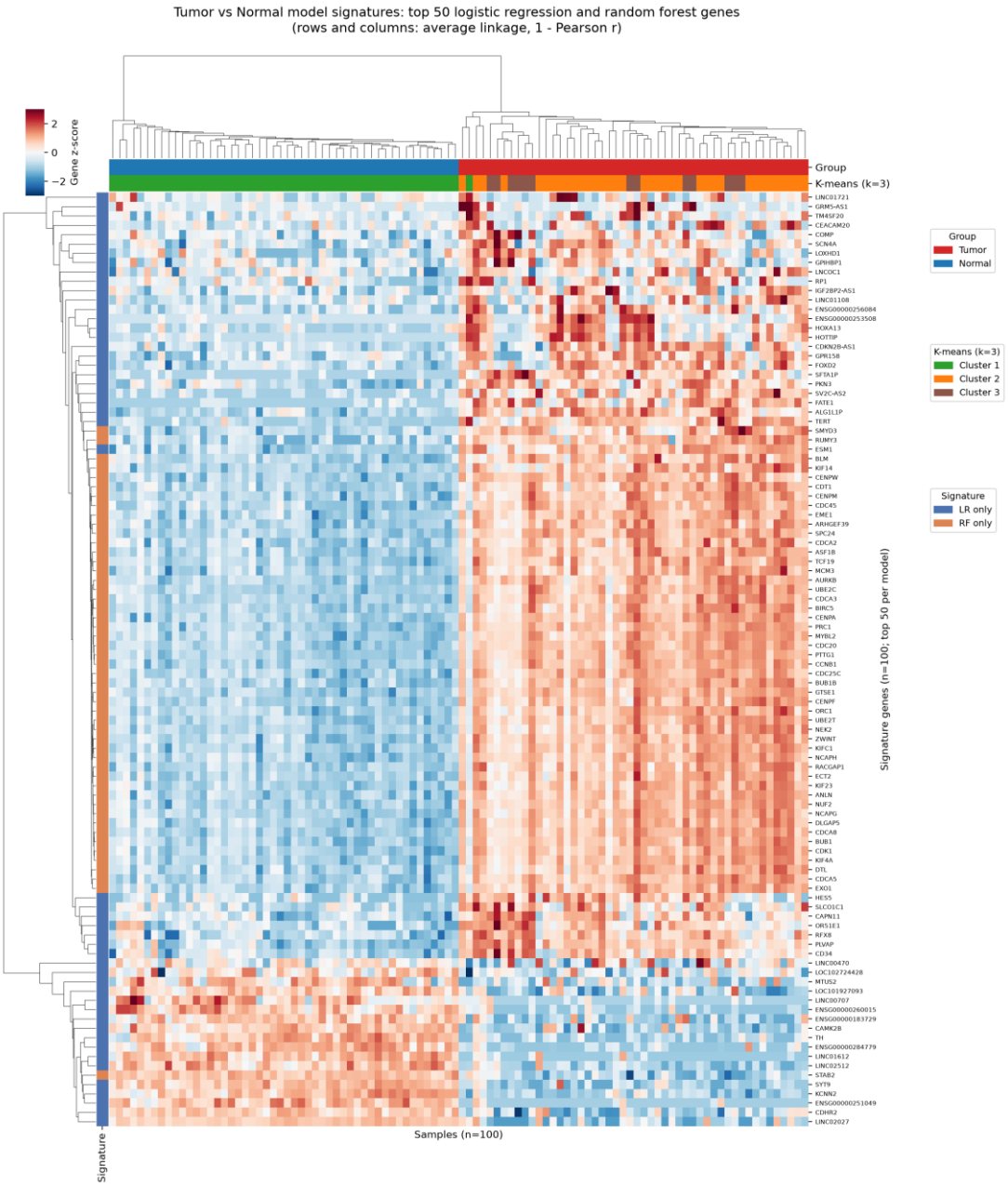


Figure 3. Signature genes of the Tumor / Normal models (rows, z-scored; top 50 per method) × 100 samples (columns), with row and column dendrograms; annotation bars show Tumor / Normal and the K-means (k = 3) clusters, the row bar shows which model selected each gene.

To define the gene signatures, both Tumor / Normal models were refitted on all 100 samples with the 5,000 most variable genes. The logistic regression signature is the 50 genes with the largest absolute coefficients and the random forest signature is the 50 genes with the highest impurity importance. The union (100 genes) was drawn as a heatmap of

per-gene z-scores, with rows and columns clustered by average linkage on correlation distance (1 − Pearson r), as in Assignment 3.

This heatmap contains only the 100 genes in either model signature; the two signatures do not share a single gene. The column dendrogram splits the samples into exactly 50 Normal and 50 Tumor, and Patient 39 Tumor sits at the edge of the tumor branch – unlike the classifiers, which use all 5,000 genes and call it Normal. The random forest genes form one large block of proliferation and cell-cycle genes (CDK1, BUB1, CCNB1, AURKB, CENPA) that are uniformly higher in tumors, whereas the logistic regression genes include tumor-high genes such as TERT, HOXA13 and HOTTIP that are strong only in subsets of tumors, plus a block of normal-high genes (e.g. CDHR2, TH, MTUS2) and several lncRNAs. The difference likely reflects the two methods: the L2 penalty spreads weight over many correlated proliferation genes, so each receives a small coefficient, while random forest splits repeatedly on the cleanest single separators.

The highest-ranked genes are LOC102724428, MTUS2 and TERT for logistic regression and ARHGEF39, ANLN and CENPA for random forest; the full signature with both weights is in signature_genes.csv in the repository. All 100 signature genes except one are significant in the Assignment 2 DESeq2 analysis (padj < 0.05), compared with 73.6% of the 5,000 candidate genes, and 86% pass the strict |log2FC| > 2 cut-off, compared with 31.7% of candidates. For 99 of the 100 genes the sign of the logistic regression coefficient agrees with the direction of the DESeq2 fold change, so the supervised models recover the same differential expression signal found in Assignment 2, with random forest focusing on the proliferation program that dominated the Assignment 2 enrichment results.

Conclusion

Both supervised methods predict Tumor vs Normal almost perfectly on held-out patients (AUC 0.999–1.000 with 5,000 genes), and even 10 genes give an AUC above 0.99. Only Patient 39 Tumor is consistently misclassified, matching its placement with normal liver in Assignment 3. The Assignment 3 clusters can also be learned, but the tumor subclusters at k = 3 need many more genes and their prediction stability is largely independent of Tumor / Normal stability, which suggests they reflect weaker within-tumor structure. Performance increases with the number of genes up to about 1,000 genes and then plateaus. The signature genes are almost all Assignment 2 DEGs, centred on cell-cycle and proliferation genes, which again answers our scientific question: gene expression profiles reliably distinguish HCC tumor tissue from matched normal liver.